

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2023-24

Paper No. BSC-07A: Ancillary Physics-II

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. Answer any **five** of the following: [5 × 4 = 20]

- (a) Define the interference and diffraction of light with suitable diagrams.
- (b) On which law of thermodynamics is a thermometer based? Explain your answer.
- (c) Discuss polarization by reflection with a suitable diagram.
- (d) How do you measure specific heat capacity? Why does sand heat up faster than water on beaches?
- (e) Explain one biological process that uses a law of thermodynamics.
- (f) Define the entropy of a system. In which thermodynamic process does the entropy not change and why?
- (g) Write the expression for the focal length and power of a thick lens of thickness t and radii of curvature R_1 and R_2 .
- (h) Consider a bacterium of mass 10^{-12} g. Find the mass of one mole of bacteria in units of elephant masses (assume an average elephant mass of 6000 kg).
- (i) Derive the ideal gas equation by using the empirical laws for gases. State one application of the ideal gas equation.

Question 2.

- (a) Explain the diffraction pattern due to a straight edge with a suitable diagram and write the expressions for maximum and minimum intensities. [6]
- (b) Define polarization with a suitable diagram and discuss the difference between longitudinal and transverse waves. [8]

Question 3.

- (a) Discuss the differences between a telescope and a microscope and their respective applications. [4]
- (b) What is the refractive index of a medium? The absolute refractive indices of glass and water are $3/2$ and $4/3$, respectively. If the speed of light in glass is 2×10^8 m/s, calculate the speed of light in air and water. [5]

- (c) Define Fresnel and Fraunhofer diffraction with suitable diagrams. [5]

Question 4.

- (a) Explain Carnot's cycle using a schematic diagram. A Carnot engine operates between reservoir temperatures of 800 K and 300 K. Calculate the percentage efficiency of the engine. [9]
- (b) What is Brownian motion? Prove that heavier molecules exhibit slower Brownian motion. [5]

Question 5.

- (a) Helmholtz free energy is given by $F = U - TS$, where U is internal energy, T is temperature and S is entropy. Deduce the expressions for entropy and pressure as negative rates of change of the Helmholtz function at fixed volume and temperature, respectively. [6]
- (b) State the first law of thermodynamics. Apply it to ideal gas expansion and comment on the result. [6]

OR

- (a) Show that when unpolarized light passes through a polarizer, the intensity of the transmitted light is exactly half of the incident light intensity. [6]
- (b) What is a diffraction grating? If the width of each slit is b and the separation between the two slits is a , write the expression for the intensity distribution and draw the diffraction pattern. [6]